

WASP-43b

R-2599

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140213 · created 2026-08-28T00:20:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456701.89702 +/- 0.00067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.015
Transit depth (Rp/Rs)^2	2.13 +/- 0.43 [%]
Orbital Inclination (inc)	83.52 +/- 1.72
Airmass coefficient 1 (a1)	1.0 +/- 0.011
Airmass coefficient 2 (a2)	0.002 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#3 - [244, 90]
Optimal Method	PSF photometry
Transit Duration (day)	0.0536 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.146 ± 0.015 vs 0.1594 (deviation 0.89σ)
Duration fit vs published	0.0536 d vs 0.0512 d (deviation 0.56σ)
Mid-time O-C	-1.9 min
Depth SNR	9.7
Residual scatter	1.12 %

Light curve

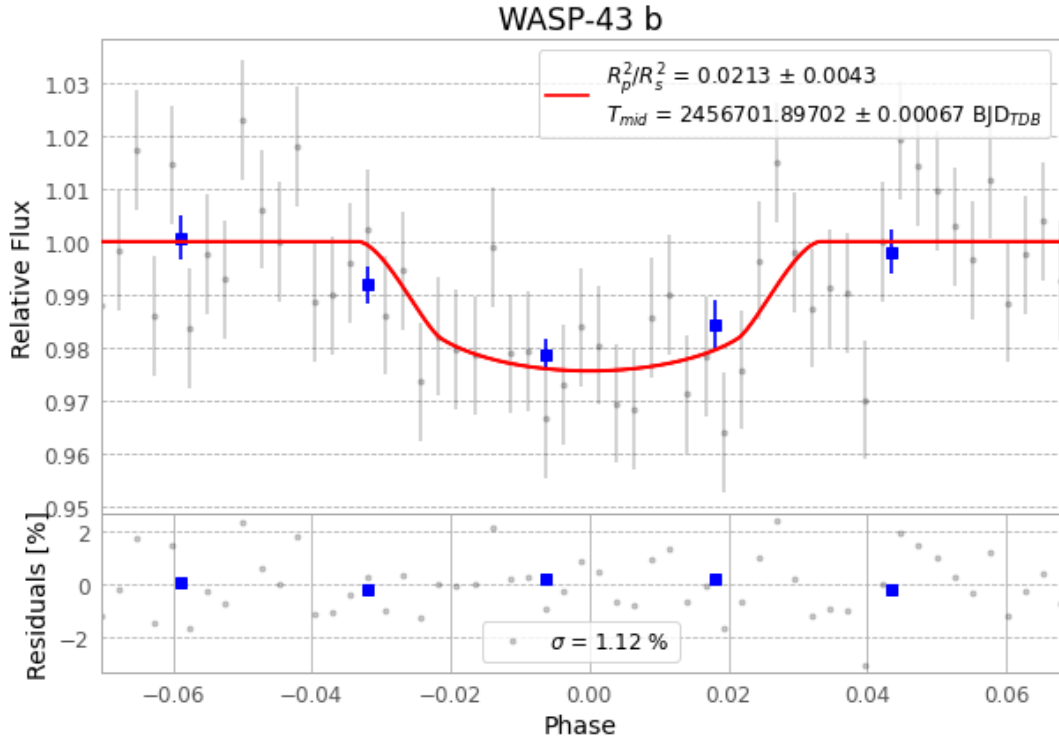


Figure 1 — FinalLightCurve_WASP-43 b_2014-02-13.png (SHA-256 ba3c357a9b1240a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [330, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f5f3230e0b289e27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), WASP-43140213080012.FITS → WASP-43140213104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140213.log (5bc21a05a038...), FinalLightCurve_WASP-43 b_2014-02-13.png (ba3c357a9b12...), FinalLightCurve_WASP-43 b_2014-02-13.csv (c99f6348fae7...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:20Z.